Reference solution 2：

For high-energy electrons,

$$v \approx c\,, \tag{1}$$

$$E = mc^2\,, \tag{2}$$

$$a = \frac{v^2}{R}\,, \tag{3}$$

$$F = ecB\,, \tag{4}$$

$$F = ma\,, \tag{5}$$

These conditions give the electron's bending radius:

$$R = \frac{E}{ecB}\,. \tag{6}$$

The number of electrons in the storage ring is

$$n = \frac{Q}{e} = \frac{I\Delta t}{e}\,, \tag{7}$$

where Δt is one revolution time. From (3) and the radiation condition, an electron loses a total energy per turn given by (there is no radiation on the straight sections)

$$E_{\text{turn}} = \frac{e^2 a^2 \gamma^4}{6\pi c^3 \varepsilon_0} \times \frac{2\pi R}{c} = \frac{e^2 \gamma^4}{3\varepsilon_0 R}\,. \tag{8}$$

Equations (7)、(8) give the total power dissipated by the ring electrons:

$$P_{\text{total}} = \frac{nE_{\text{turn}}}{\Delta t} = \frac{Ie\gamma^4}{3\varepsilon_0 R} = 4.34\times 10^5\,\text{W}\,. \tag{9}$$

The power at each outlet is

$$P_{\text{O}} = \frac{P_{\text{total}}}{N} = 2.17\times 10^4\,\text{W}\,. \tag{10}$$